Laboratory 03 Report: PCA and LDA

Machine Learning and Pattern Recognition

1 Project section

2 PCA analysis

What do you observe from the PCA projected histograms? What are the effects on the class distribution?

Applying Principal Component Analysis (PCA) transforms the dataset by projecting it onto a new set of orthogonal axes ordered by their variance. Observing the histograms of the projected features, the first PCA direction (Direction 1) captures the maximum overall spread of the dataset. As we move towards Direction 6, the variance noticeably decreases.

However, the effect on the class distribution reveals a crucial limitation of unsupervised dimensionality reduction. Both the "Fake" (label 0) and "Genuine" (label 1) classes span across the entire range of these new axes and heavily overlap. Because PCA maximizes the total variance of the data regardless of class labels, the direction of maximum data spread does not align with the direction of maximum class discrimination. The distributions morph into largely unimodal shapes, but they do not separate.

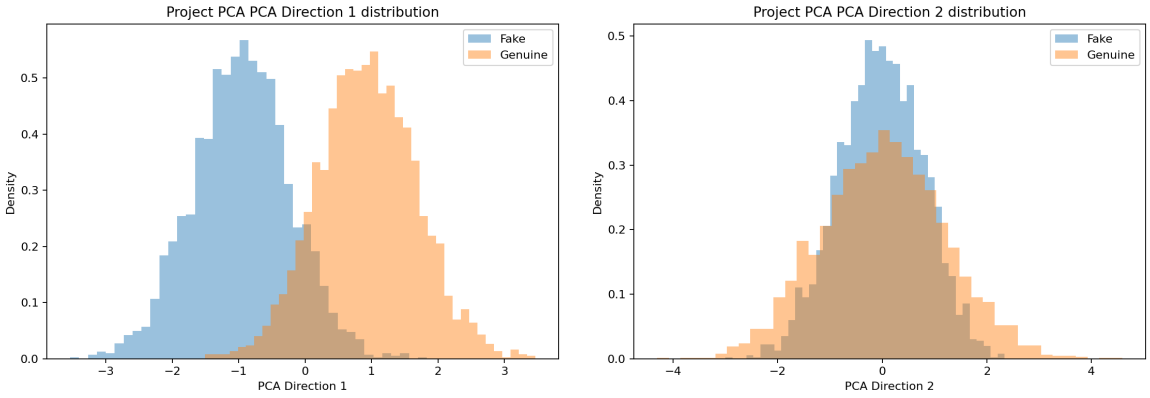


Figure 1: Histograms of the data projected onto the first two PCA directions. Notice the heavy overlap between the Fake and Genuine classes despite the high overall variance.

Can you spot the different clusters inside each class?

Examining the pairwise scatter plots of the PCA-transformed features, we look for distinct, isolated sub-populations (clusters) within the individual classes. The scatter plots show that both the "Fake" and "Genuine" classes form dense, continuous, and somewhat elongated clouds. There are no visually distinct or separated sub-clusters within either class. The structural variations captured by the principal components do not break the classes apart into isolated islands.

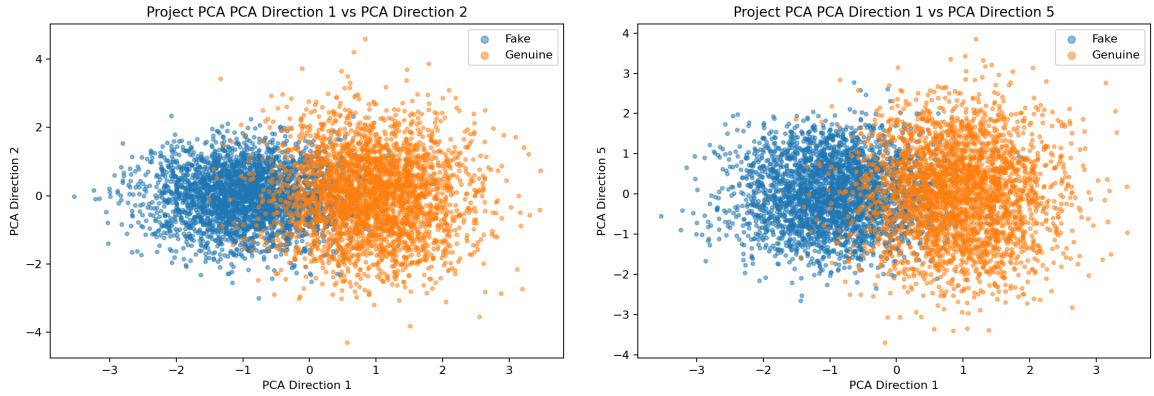


Figure 2: Scatter plots of PCA Direction 1 vs 2 (left) and Direction 1 vs 5 (right). The classes form single, continuous clouds without isolated internal sub-clusters.

Have the clusters really changed? Have the classes really become "more separated"?

No, the clusters have not changed, and the classes have not become more separated.

Applying PCA and retaining all $m = 6$ dimensions acts mathematically as a rigid orthogonal rotation of the original feature space. Because an orthogonal transformation perfectly preserves Euclidean distances (L_2 norm), any two data points that were overlapping or close together in the raw feature space remain exactly the same distance apart in the PCA-transformed space. We are simply viewing the exact same geometric structure from a newly rotated coordinate system.

This confirms that exploratory analyses (like viewing 1D and 2D shadows of a 6D object) are often insufficient to understand global characteristics. To actually increase separability, a supervised approach is required. For instance, applying Linear Discriminant Analysis (LDA) specifically targets the direction that maximizes the distance between class means while minimizing intra-class variance. As seen below, the LDA projection successfully reveals a distinct shift between the classes that PCA's blind rotation could not achieve.

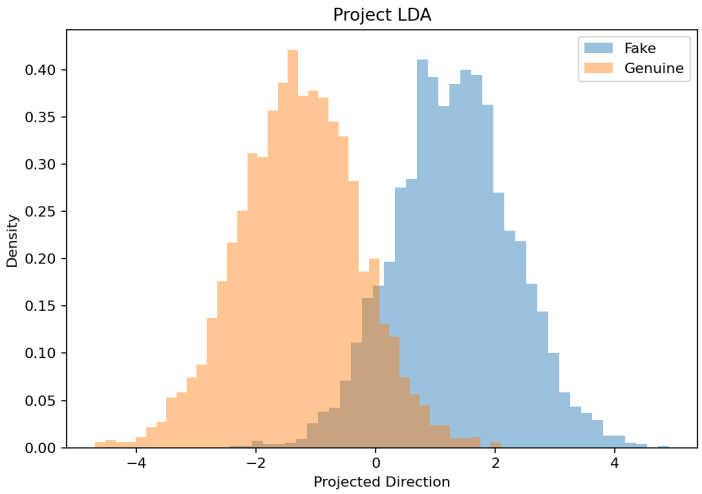


Figure 3: LDA Projection (1D). Unlike PCA, the supervised LDA projection clearly separates the density centers of the Fake and Genuine classes.

3 LDA analysis

Apply LDA and compute the histogram of the projected samples. What do you observe? Do the classes overlap over the found direction?

Observing the histogram of the samples projected onto the single LDA direction, we see a stark contrast to unsupervised methods like PCA. The data forms a clearly bimodal distribution with the "Fake" and "Genuine" classes occupying distinct regions of the projected space. However, the classes do still overlap over the found direction. The right tail of the "Fake" class intersects with the left tail of the "Genuine" class in the center. This indicates that while linear separability is high, a perfectly clean linear boundary that yields 0% error does not exist for this dataset.

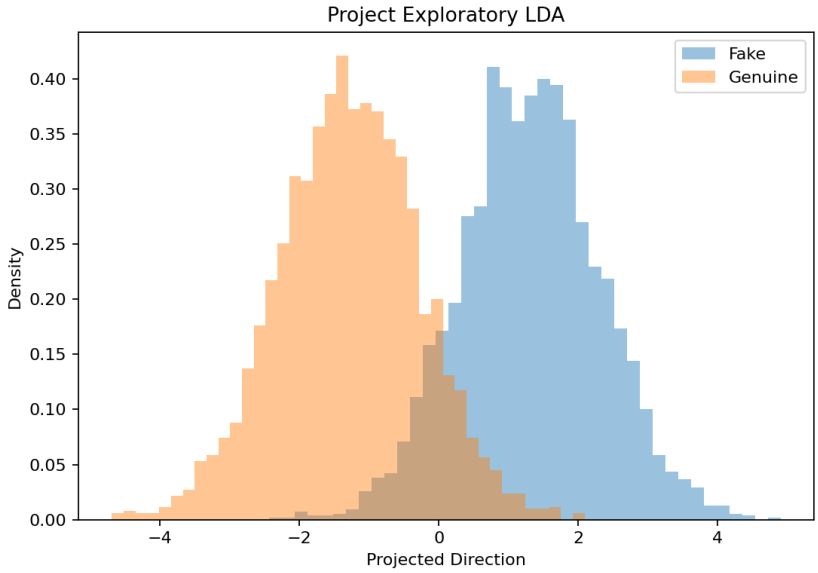


Figure 4: Histogram of the data projected onto the LDA direction. The density centers are separated, but the tails overlap.

Compared to the histograms of the 6 features you computed in Laboratory 2, is LDA finding a good direction with little class overlap?

Yes, LDA is finding an exceptionally good direction compared to the raw features. In Laboratory 2, the histograms of the 6 individual features showed massive overlap; the distributions for "Fake" and "Genuine" were nearly superimposed, meaning no single raw feature was a strong discriminator. LDA has successfully found a linear combination of these original features—an optimal viewing angle—along which the classes were already well separated, drastically reducing the overlap volume compared to any single raw axis.

Compute the predictions on the validation data, and the corresponding error rate.

By splitting the dataset, training the LDA projection on the training set, and applying it to the validation set, we establish a baseline linear classifier. We oriented the projection so that the mean of the "Genuine" class is larger than the "Fake" class. Using the default threshold—calculated strictly as the geometric average of the two projected training class means (approx 0.015)—the model achieved an error rate of **9.57%** on the validation data.

Now try changing the value of the threshold. What do you observe? Can you find values that improve the classification accuracy?

By sweeping the threshold across the range of projected values and evaluating the validation error at each step, we observe that the error rate forms a distinct curve with a clear minimum. The default threshold (the mean of the projected classes) does not yield the absolute lowest error.

We can indeed find values that improve classification accuracy. By shifting the threshold slightly higher (to approximately 0.021), the validation error rate improves, dropping from 9.57% down to **8.74%**. This observation suggests that the two class distributions are not perfectly symmetric or do not share the exact same variance; therefore, a slightly biased decision boundary discriminates the overlapping tails more effectively than the exact geometric midpoint.

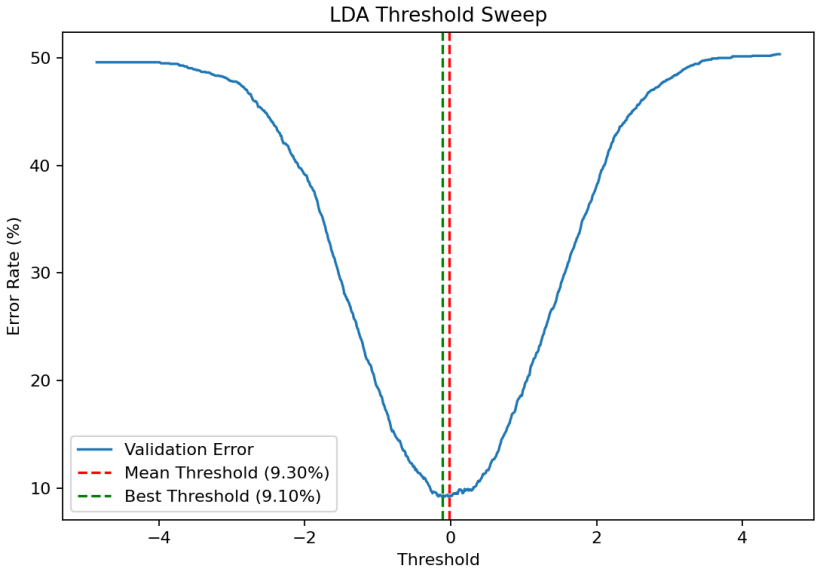


Figure 5: Validation error rate as a function of the classification threshold. The green dashed line indicates the empirical optimum, which achieves a lower error rate (8.74%) than the default geometric mean threshold (9.57%, red dashed line).

Apply PCA as a pre-processing step for LDA. Evaluate the error rate for different values of PCA dimensions ($m = 1, \ldots, 6$). What do you observe?

To evaluate the impact of PCA as a pre-processing step, we first project the data onto an m-dimensional PCA subspace, and subsequently apply LDA to reduce it to a single dimension for classification. The default threshold (average of the projected means) is used for evaluation on the validation set.

As shown in Table 1, when we keep all dimensions ($m = 6$), the error rate perfectly matches the baseline pure LDA model. This is mathematically expected, as keeping all principal components simply acts as a rigid rotation of the feature space without losing any information.

However, as we reduce the number of PCA dimensions (m), the error rate typically degrades (increases).

PCA Dimensions (m)	**Validation Error Rate (%)**
1	*[Insert CSV value for m=1]*
2	*[Insert CSV value for m=2]*
3	*[Insert CSV value for m=3]*
4	*[Insert CSV value for m=4]*
5	*[Insert CSV value for m=5]*
6	9.57

Table 1: Classification error rates using PCA as a pre-processing step for LDA across different subspace dimensions.

Does PCA pre-processing improve the classification accuracy? Why or why not?

No, PCA pre-processing does not improve the classification accuracy for this dataset; in fact, aggressively reducing dimensions harms it.

This behavior highlights a fundamental difference between the two algorithms. PCA is an unsupervised technique that discards directions with the lowest overall variance. However, variance does not necessarily equate to class separability. A feature might have very little variance overall, but if that slight variance is entirely due to the difference between the "Fake" and "Genuine" classes, it is highly discriminative. By dropping these "low variance" directions, PCA is inadvertently throwing away information that LDA could have used to separate the classes. Thus, for this specific dataset, feeding the raw features directly to LDA yields the best linear separation.